

Geometry–Light Relations in Modal Triplet Theory: Exact Identities, Conditional Bounds, and Principal Symbols

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Abstract

We separate the geometry-light statements of MTT into exact algebraic identities, consequences of explicit symmetry or positivity assumptions, and phenomenological estimates requiring additional dynamics. Modal democracy implies $\sin^2 \theta_W = 3/8$, but democracy is an assumption and is not selected by the current gauge profile. Holonomy phase sums require a specified trivial tensor product and compatible connection. Equal propagation speeds require equality of principal symbols and do not follow from topology alone. Curvature–mass drift is an identity conditional on a chosen mass–curvature ansatz, while the PPN statement requires an explicit response-norm estimate. No internal spectral gap is identified with an external cutoff. These typed statements provide safe interfaces for later numerical tiers without overstating their physical content.

Revision note for this edition

Supersedes. *Geometry–Light Relations in Modal Triplet Theory*, version 2.

Reason. Algebraic identities, symmetry assumptions, topology, spectral gaps, propagation speeds, and phenomenological bounds were not consistently separated by logical strength.

Resolution. Version 3 types each statement as exact, conditional, or phenomenological, moves causal speed to principal symbols, and separates internal gaps from external cutoffs.

Retained result. Modal-democracy, holonomy, curvature–mass, and PPN relations remain valid in precisely the displayed conditional forms.

Remaining boundary. Physical light propagation and quantitative bounds require a selected hyperbolic action, response map, and normalization.

1 Classification rule

A relation is called *exact* only when it follows algebraically from its displayed definitions. A *conditional theorem* additionally lists every symmetry, bundle, principal-symbol, positivity, or response assumption. A *phenomenological bound* also requires a selected physical normalization and comparison scheme. Geometry-light does not mean assumption-free.

2 Gauge-weight identity and modal democracy

At one common scale and in one scheme, write

$$\alpha_r^{-1} = K\zeta_r, \quad \zeta_r > 0,$$

and use $g_Y^2 = (3/5)g_1^2$. Then the exact algebraic identity is

$$\boxed{\sin^2 \theta_W = \frac{1}{1 + (5/3)(\zeta_1/\zeta_2)}}. \quad (2.1)$$

Definition 2.1 (Modal democracy). Modal democracy at the chosen scale means $\zeta_1 = \zeta_2$.

Corollary 2.2 (Conditional democracy value). *If modal democracy holds, Equation (2.1) gives $\sin^2 \theta_W = 3/8$.*

This is exact given democracy, but democracy is not derived by this paper and is not satisfied by the selected low-scale SMDR profile. The value $3/8$ is therefore neither a current MTT weak-angle prediction nor evidence for a physical matching scale. For $\zeta_1/\zeta_2 = 1 + \varepsilon$, the Taylor identity is

$$\sin^2 \theta_W = \frac{3}{8} \left(1 - \frac{5}{8}\varepsilon + \frac{25}{64}\varepsilon^2 + O(\varepsilon^3) \right).$$

3 Conditional holonomy phase sum

Theorem 3.1 (Holonomy product under compatible trivialization). *Let L_{12}, L_{23}, L_{31} be line bundles equipped with unitary connections. Assume a specified connection-preserving trivialization*

$$L_{12} \otimes L_{23} \otimes L_{31} \cong \underline{\mathbb{C}}.$$

Then for every closed loop γ ,

$$U_{12}(\gamma)U_{23}(\gamma)U_{31}(\gamma) = 1,$$

and the three phases sum to an integer multiple of 2π .

Proof. The tensor-product connection is trivial by assumption, so its holonomy is unity. Holonomy on a tensor product is the product of the three line-bundle holonomies. $\square$

Topological triviality alone does not force an arbitrary chosen product connection to have zero holonomy. The compatible trivialization is an essential hypothesis. The theorem constrains phase sums but does not select CKM or PMNS phases individually.

4 Principal-symbol condition for propagation speeds

For a linearized field component ϕ_a , let the second-order operator have principal symbol

$$\sigma_2(P_a)(x, \xi) = G_a^{\mu\nu}(x)\xi_\mu\xi_\nu.$$

Theorem 4.1 (Common characteristic cone). *Two canonically normalized sectors have the same local high-frequency propagation cone if and only if their nondegenerate principal symbols define the same null cone. In dimension at least three this means $G_b^{\mu\nu} = \Omega^2 G_a^{\mu\nu}$ for a positive conformal factor, subject to the usual regularity assumptions.*

Lower-order masses, potentials, projectors, or internal overlap coefficients do not determine the characteristic cone. Equal wave speeds therefore require a common principal-symbol theorem; they cannot be inferred solely from bundle topology, modal counting, or an internal spectral gap.

5 Representation-only RG signs

Conditional on importing the Standard Model field content and perturbative quantization, the one-loop coefficient

$$b_0 = \frac{11}{3}C_A - \frac{4}{3}\sum_f T(R_f) - \frac{1}{6}\sum_s T(R_s)$$

fixes the qualitative signs of the gauge beta functions. This is a standard QFT consequence of representation content, not a new geometric prediction of MTT. Quantitative running uses the selected multi-loop transport rather than this sign-level approximation.

6 Conditional curvature–mass drift

Assume, rather than derive, the local ansatz

$$m(x) = \sqrt{\kappa(\lambda^{(0)} + \beta R(x))}, \quad \lambda^{(0)} + \beta R(x) > 0.$$

Then differentiation gives the exact identity

$$\nabla_\mu \log m = \frac{\beta \nabla_\mu R}{2(\lambda^{(0)} + \beta R)}.$$

If the denominator is bounded below by $\lambda_{\min} > 0$, then

$$|\nabla_\mu \log m| \leq \frac{|\beta|}{2\lambda_{\min}} |\nabla_\mu R|.$$

The calculus is exact, but the mass ansatz, β , and physical-unit bridge are independent inputs. This is not a prediction of cosmological mass drift.

7 Conditional PPN estimate

Suppose a linearized gravitational equation can be written

$$\square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu} + S_{\mu\nu}$$

and a specified solution norm obeys

$$\|S\| \leq C\Delta_{\text{curv}}\|T\|.$$

Only under a stable inverse estimate for the gauge-fixed operator does this imply a bound of the form

$$|\gamma - 1| \leq C_{\text{PPN}}\Delta_{\text{curv}}.$$

The constants and norm must be computed for the selected background and source. Writing merely $\gamma = 1 + O(\Delta_{\text{curv}})$ is an asymptotic template, not a numerical solar-system prediction.

8 No internal-gap cutoff inference

An internal Laplacian gap controls suppression of omitted internal modes in a specified reduction. It does not by itself define a four-dimensional UV, coherence, inflationary, or cosmological cutoff. Any such identification needs a dimensionful bridge, an action-level decoupling theorem, and an error bound. The common SMDR point $Q = M_t$ is likewise a matching convention rather than a physical cutoff.

9 Conclusion

The geometry-light tier contains useful exact relations, but their scope is now explicit. The weak-angle and curvature formulas are algebraic identities conditional on stated ansatz data; holonomy phases require a compatible trivialization; wave-speed equality is a principal-symbol question; RG signs are imported Standard Model QFT; and PPN control needs a quantitative response estimate. These statements can constrain later constructions without being misreported as source-derived numerical predictions.

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